Q. Classify the input pattern shown in the figure using the hebb training algorithm. The input are "1" where "+" symbol is present and "-1" where "." is present. The "L" pattern belong to "U" pattern not belong to the class

+	+	+	+	+	+
+	.	+	+	.	.
+	+	+	+	+	+
+	.	+	+	.	.
+	.	+	+	+	+

A E

Pattern	x_1	x_2	x_3	x_4	x_5	x_6	x_7	x_8	x_9
L	1	1	1	1	-1	1	1	1	1
U	1	1	1	1	-1	-1	1	1	1

x_{10}	x_{11}	x_{12}	x_{13}	x_{14}	b	o/p	x_{15}
1	-1	1	1	-1	1	1	1
1	-1	-1	1	1	1	-1	1

Pattern 1, $w_1 = w_2 \ldots = w_{14}$

$\Delta b = y = 1$

$\Delta w_1 = x_1 y = 1$ $\Delta w_6 = x_6 y = 1$ $\Delta w_{11} = x_{11} y = -1$

$\Delta w_2 = x_2 y = 1$ $\Delta w_7 = x_7 y = 1$ $\Delta w_{12} = x_{12} y = 1$

$\Delta w_3 = x_3 y = 1$ $\Delta w_8 = x_8 y = 1$ $\Delta w_{13} = x_{13} y = 1$

$\Delta w_4 = x_4 y = 1$ $\Delta w_9 = x_9 y = 1$ $\Delta w_{14} = x_{14} y = -1$

$\Delta w_5 = x_5 y = -1$ $\Delta w_{10} = x_{10} y = 1$ $\Delta w_{15} = x_{15} y = 1$

$w_i(new) = w_i(old) + \Delta w_i \qquad b(new) = b(old) + y = 1$

$w_1(new) = 0+1 = 1$
$w_2(new) = 0+1 = 1$
$w_3(new) = 0+1 = 1$
$w_4(new) = 0+1 = 1$
$w_5(new) = 0+(-1) = -1$

$w_6(new) = 0+1 = 1$
$w_7(new) = 0+1 = 1$
$w_8(new) = 0+1 = 1$
$w_9(new) = 0+1 = 1$
$w_{10}(new) = 0+1 = 1$

$w_{11} = -1$
$w_{12} = 1$
$w_{13} = 1$
$w_{14} = -1$
$w_{15} = 1$

$b(new) = 0+1 = 1$

Pattern V

$\Delta w_1 = x_1 y = -1$
$\Delta w_2 = x_2 y = -1$
$\Delta w_3 = x_3 y = -1$
$\Delta w_4 = x_4 y = -1$
$\Delta w_5 = x_5 y = 1$

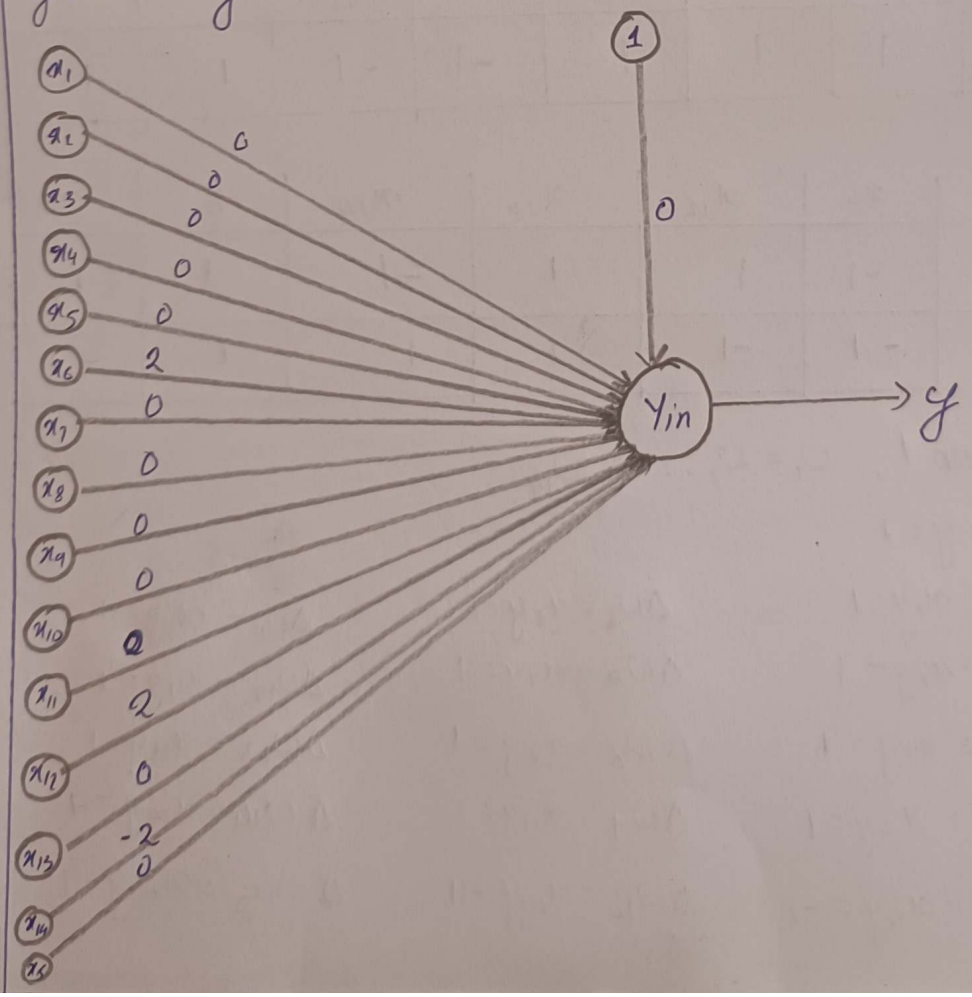
$\Delta w_6 = 1$
$\Delta w_7 = -1$
$\Delta w_8 = -1$
$\Delta w_9 = -1$
$\Delta w_{10} = -1$

$\Delta w_{11} = 1$
$\Delta w_{12} = 1$
$\Delta w_{13} = -1$
$\Delta w_{14} = -1$
$\Delta w_{15} = -1$

$b\,new = -1+1 = 0$

$w_1(new) = 1+-1 = 0$
$w_2(new) = 1+-1 = 0$
$w_3(new) = 1+-1 = 0$
$w_4(new) = 1+-1 = 0$
$w_5(new) = -1+1 = 0$

$w_6(new) = 1+1 = 2$
$w_7(new) = 1+-1 = 0$
$w_8(new) = 1+-1 = 0$
$w_9(new) = 1+-1 = 0$
$w_{10} = 0$

$w_{11} = -1+1 = 0$
$w_{12}\,new = 1+1 = 2$
$w_{13}(new) = 1+-1 = 0$
$w_{14}(new) = -1+-1 = -2$
$w_{15}(new) = 1+-1 = 0$

final weight